

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CHM 316 — Past Questions & Materials
by Renance

Course: CHM 316
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 300

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Paper 1 of 1 — CHM 316 POLYMER CHEMISTRY 2024/2025

Level 300 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF CHEMISTRY

FIRST SEMESTER 2024/2025 EXAMINATION

CHM 316: POLYMER CHEMISTRY (2 CREDITS)

Instructions: Attempt All Questions in both sections Time: 1'4 hours

SECTION A (30 marks)

- 1a) A polymeric sample in which 30% molecules have a molecular mass of 20,000, 40% have 30,000 and the rest 30% have 60,000. Calculate the number-average molecular weight and the weight-average molecular weight of the polymeric sample. (5 marks)
- b) Name and draw the structures of three condensed polymers and three addition polymers, identify the basic monomeric structures. (3 marks)
- c) Name six polymers that you encounter daily (3 marks).
- d) Name one polymer for each of the following applications: (i) For insulating electric wires. (1 mark)
For non-stick coatings on kitchen utensils. (1 mark)
(ii) For blending with cotton to make shrink-resist cloth (1 mark)
(iii) What two polymers are used to make soft-drink bottles and are often recycled? (2 marks)
- e) Is $[-CH_2=CH(C_6H_5)-]_n$ a homopolymer or copolymer? Write the name of monomer. (2 marks)
- f) What are the three main stages of free-radical polymerization? (3 marks)
- g) Name the functional group(s) in each of these monomers: (13 marks)
(i) adipic acid. (3 marks)
(ii) styrene (3 marks)
(iii) hexamethylenediamine (3 marks)
(iv) acrylonitrile (3 marks)
(v) ethene (3 marks)
- h) Using poly (acrylonitrile) as an example, draw a portion of a linear polymer, a branched polymer and a random copolymer when polymerised with ethene. (3 marks)
- i) Discuss the mechanism of free-radical polymerization using a schematic sequence only. (3 marks)
- j) Draw the structures of the following polymers. In each case, indicate the type of polymerization. (3 marks)
(i) styrene (ii) hexamethylenediamine (iii) acrylonitrile (iv) ethene (v) adipic acid
- 7) Given that A and B are two monomers, write generic structures of alternate, block and graft copolymers. (3 marks)

SECTION 2B (40 marks)

the following set of monomers is polymerised:

a)

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You are given

Sample ~ | Number of moles | Molecular weight |

Sample 1 | 0.0001 mol, 1000 g |

Sample 2 | 0.0002 mol, 2000 g =

Sample 3 | 0.0001 mol, 1000 g

Sample 4 | 0.0003 mol, 3000 g

Sample 5 | 0.0001 mol, 1000 g

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(a) Calculate the number and weight averages of the mixture of these set of samples, (10 marks)

(b) What is the polydispersity index of the mixture? (4 marks)

(c) State two differences between addition and condensation polymerization. (6 marks)

(i) Cellulose (11)

(ii) Which of the following polymers are natural and which are synthetic?

acrylonitrile)

Polystyrene (1) Polyethylene (iv) Starch (v) Proteins (vi) Silicones (vii) Polysiloxanes

(viii) Phenol-formaldehyde resins (ix) Wool (x) Poly(vinyl acetate) (xi) Plastics (xii) Natural

rubber (xiii) Cotton (xiv) Nylon (xv) Teflon. (8 marks)

(t) Explain in detail the mechanism of free radical polymerization of acrylonitrile using peroxide as initiator. (10 marks)

(c) What is the main difference

reactions? (2 marks)

benzoin

between free-radical, cationic and anionic polymerization

END